Black body radiation (1900) by Max Planck : origin of QM

Black body = object that absorbs all radiation incident upon it
no reflection

When heated, it emits radiation.

In thermal equilibrium, black body emits and absorbs radiation at the same rate.

Suppose oven with walls at temperature $\tau = k_B T$.
(Josef Stefan) found in 1879 that the rate of radiation coming out via small hole of area A is

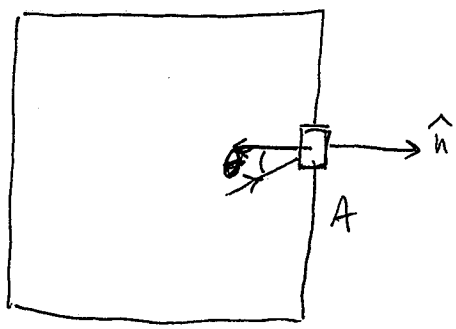
$$\frac{dQ}{dt} = A\sigma T^4$$

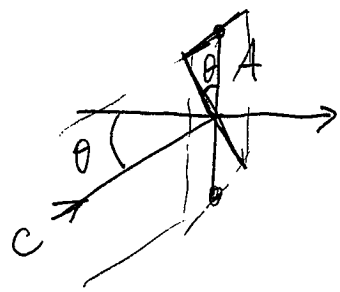
Stefan constant $\sigma = 5.67 \cdot 10^{-8}\ W\ m^{-2} K^{-4}$

Naturally, $\frac{dQ}{dt} \sim$ energy density u
(energy in a unit volume of cavity)

Spectral density of radiation $u(\lambda)\,d\lambda =$
amount of energy in unit volume in the range $\lambda, \lambda + d\lambda$ of <u>wavelength</u>

Radiation travels with speed of light c





Volume $c\,dt\,A\cos\theta$

Energy in the wavelength range <u>λ to $\lambda + d\lambda$</u>

$$c\,dt\,A\cos\theta \cdot u(\lambda)\,d\lambda \underbrace{\frac{d\theta \sin\theta\, d\varphi}{4\pi}}$$

measure of integration = fraction of incoming waves whose direction lies between θ and $\theta + d\theta$ and φ and $\varphi + d\varphi$

$\pi/2$ ← integrate over ½ space!

$$\Rightarrow \frac{dQ}{dt} = \frac{cA}{4\pi}\, u(\lambda)\,d\lambda \underbrace{\int_0^{2\pi} d\varphi}_{2\pi} \underbrace{\int_0^{\pi/2} d\theta \sin\theta \cos\theta}_{-\int_1^0 d\cos\theta \cos\theta = \frac{1}{2}}$$

summing over all wavelength

$$\Rightarrow \frac{dQ}{dt} = \frac{Acu}{4} \quad \text{with } u = \int_0^\infty u(\lambda)\,d\lambda$$

$$\Rightarrow A\sigma T^4 = \frac{Acu}{4} \Rightarrow \boxed{u = \frac{4\sigma T^4}{c}}$$

follows from Stefan Boltzman law

Wilhelm Wien (1893) showed that

$$u(\lambda)\,d\lambda = \frac{f(\lambda T)\,d\lambda}{\lambda^5}$$

Wien's scaling law

$f(x)$ – unknown scaling function

$$\Downarrow$$

$$\frac{u(\lambda)}{T^5} = \frac{f(\lambda T)}{\lambda^5 T^5}$$

$\Rightarrow$ if we plot $\dfrac{u(\lambda)}{T^5}$ as a function of (λT),

all experimental data should fall onto a single curve! [Scaling]

Then the energy density

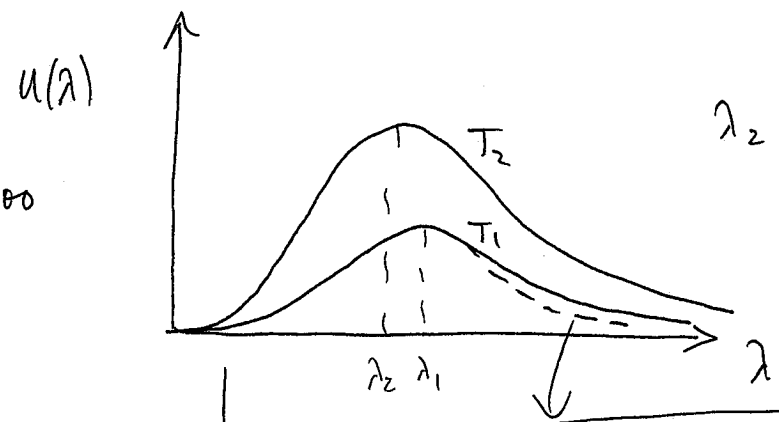
$$u = \int_0^\infty u(\lambda)\, d\lambda = \int_0^\infty \frac{f(\lambda T)\, d\lambda}{\lambda^5}$$

Max of $u(\lambda)$ plot: $\dfrac{du(\lambda)}{d\lambda} = -\dfrac{5f(\lambda T)}{\lambda^6} + \dfrac{f'(\lambda T)\cdot T}{\lambda^5}$

$$\Rightarrow \quad \lambda T\Big|_{max} = \frac{5f(\lambda T)}{f'(\lambda T)} = \lambda_{max}\, T$$

Experimentally, $\lambda_{max} T = 0.002899$ meter $\cdot$ K
(Wien's displacement law) $\rightarrow$ position of max of $u(\lambda)$, λ_{max}, shifts to shorter wavelengths as the temperature T of the black body increases.

data 1900

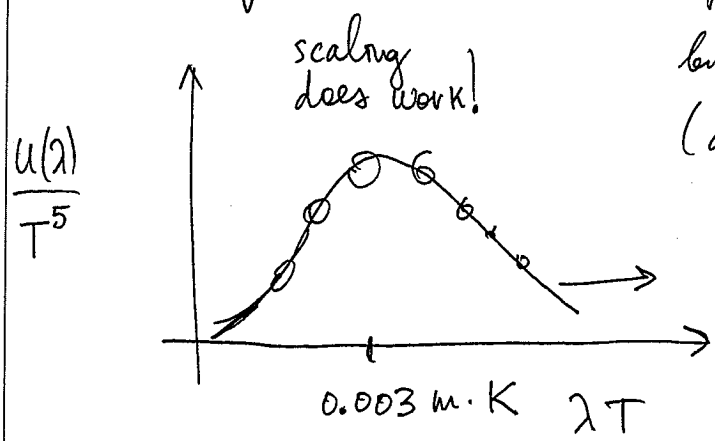


$\lambda_2 T_2 = \lambda_1 T_1$

$\lambda_2 = \lambda_1 \dfrac{T_1}{T_2} < \lambda_1$ for $T_2 > T_1$

$$\boxed{\text{Wien's proposal } u(\lambda) = \frac{c_1 e^{-c_2/\lambda T}}{\lambda^5}} \quad (*)$$

works for short λ
but not long ones
(dashed line)



$\dfrac{f(\lambda T)}{(\lambda T)^5}$ is obtained in this way.

Rayleigh – Jeans theory (1905) + Einstein (photoelectric paper (1905))

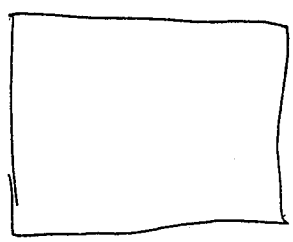
Think of E-M radiation in the cavity as being due to oscillators in the walls.

Classical E&M does say that accelerating electric charges emit ~~E&M~~ waves!

By equipartition theorem each oscillator has energy $\underbrace{\frac{1}{2}k_B T}_{(K.E.)} + \underbrace{\frac{1}{2}k_B T}_{(P.E.)} = k_B T$

$\Downarrow$

Each standing wave has energy $\underline{k_B T}$ (equilibrium!)

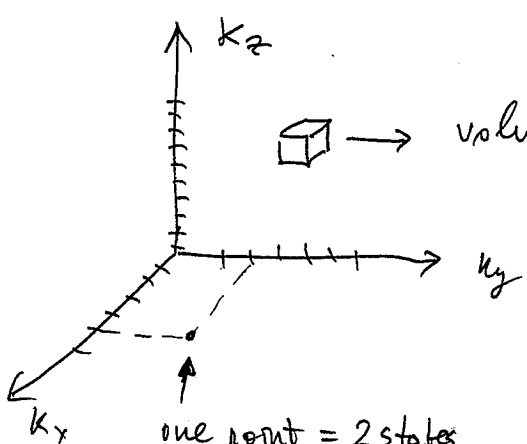


# of EM waves in the cube of length L with wavenumbers between $(k, k+dk)$

standing E-wave

$$E \sim E_0 \sin\frac{n_1 \pi x}{L_x} \sin\frac{n_2 \pi y}{L_y} \sin\frac{n_3 \pi z}{L_z}$$ (+2 transverse polarizations!)

$$\vec{k} = \frac{\pi}{L_x} n_1 \hat{x} + \frac{\pi}{L_y} n_2 \hat{y} + \frac{\pi}{L_z} n_3 \hat{z}$$



one point = 2 states (because of 2 polarizations)

$\longrightarrow$ volume of <u>one</u> unit cell in $\vec{k}$ space $\dfrac{\pi^3}{L_x L_y L_z} = \dfrac{\pi^3}{\text{Volume}} = \dfrac{\pi^3}{V}$

$\Downarrow$

the <u>density</u> of points in $\vec{k}$-space is $= \dfrac{V}{\pi^3}$

$\Downarrow$

density of states $= \dfrac{V}{\pi^3}$

Number dN of single-particle states in volume $dV_k = k^2 \sin\theta \, d\theta \, d\varphi \, dk$ of $\vec{k}$-space

$$dN' = dV_k \cdot \underbrace{\left(\frac{V}{\pi^3}\right)}_{\text{density}} \cdot (2)$$

↳ polarizations

\# states in spherical shell $(k, k+dk)$

$$dN = \frac{2V}{\pi^3} k^2 dk \underbrace{\int_0^{\pi/2} d\theta \sin\theta}_{\int_1^0 d(-\cos\theta) \cdot \frac{\pi}{2} = \frac{\pi}{2}} \int_0^{\pi/2} d\varphi$$

$$dN = \frac{2V}{\pi^3} \cdot k^2 dk \, \frac{\pi}{2} = \frac{V}{\pi^2} k^2 dk = D(k) \, dk$$

$$D(k) = \frac{V}{\pi^2} k^2 = \text{density of states}$$

(for E-M waves)

But $k = \frac{2\pi}{\lambda} \Rightarrow dk = -\frac{2\pi}{\lambda^2} d\lambda$

$$D(k) \, dk = \frac{V}{\pi^2} k^2 dk = \frac{V}{\pi^2} \left(\frac{2\pi}{\lambda}\right)^2 \frac{2\pi \, d\lambda}{\lambda^2} =$$

$$= \frac{V}{\pi^2} \frac{8\pi^3 \, d\lambda}{\lambda^4} = \frac{8\pi V}{\lambda^4} d\lambda = D(\lambda) \, d\lambda$$

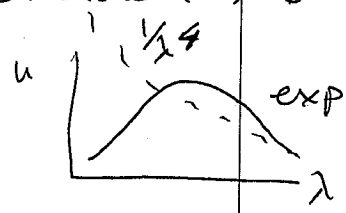
$\Rightarrow$ energy density $u(\lambda) \, d\lambda = \frac{k_B T}{V} D(\lambda) \, d\lambda = \frac{8\pi k_B T}{\lambda^4} d\lambda$

(Rayleigh – Jeans spectral energy density)

① Wien's scaling is OK: $u(\lambda) = \frac{8\pi k_B T}{\lambda^4} = \frac{f(\lambda T)}{\lambda^5}$
(but not Wien's f-la ☆)

$$\Rightarrow f(\lambda T) = 8\pi \lambda k_B T$$

But the curve $u(\lambda)$ is diverging at small λ!



Full (Total) energy density

$$U = \int_0^\infty u(\lambda)\, d\lambda = \int_0^\infty 8\pi k_B T \frac{d\lambda}{\lambda^4} =$$

$$= 8\pi k_B T \left(-\frac{1}{3}\right) \int_0^\infty d(\lambda^{-3}) = \frac{8\pi k_B T}{3} \frac{1}{\lambda^3}\Big|_{\lambda=0} = \underline{\underline{\infty!}}$$

Ehrenfest: ultraviolet catastrophe
(short λ)

<u>Planck</u> constructed a formula to satisfy both limits

$$u(\lambda) = \frac{8\pi hc}{\lambda^5 \left(e^{hc/\lambda k_B T} - 1\right)}$$

$\lambda \to 0$: $u(\lambda) \approx \frac{8\pi hc}{\lambda^5} e^{-\frac{hc}{\lambda k_B T}}$ agrees with

Wien's proposal $u(\lambda) \sim \frac{c_1}{\lambda^5} e^{-c_2/\lambda T}$

Here $c_1 = 8\pi hc$

$c_2 = \frac{hc}{k_B}$

h = new constant introduced by Planck

$$\lambda \to \infty: \quad u(\lambda) \approx \frac{8\pi hc}{\lambda^5 \frac{hc}{\lambda k_B T}} = \frac{8\pi k_B T}{\lambda^4}$$ (Rayleigh-Jeans f-la!)

Total energy density:

$$u = \int_0^\infty \frac{8\pi hc}{\lambda^5 (e^{hc/\lambda k_B T} - 1)} d\lambda = \left(z = \frac{hc}{\lambda k_B T} \right) =$$

$$dz = \frac{hc}{k_B T} \frac{-d\lambda}{\lambda^2}$$

$$= 8\pi hc \left(\frac{hc}{k_B T}\right)^{-1} \int_\infty^0 \frac{-dz \cdot z^3}{e^z - 1} \left(\frac{k_B T}{hc}\right)^3 =$$

$$= 8\pi hc \left(\frac{k_B T}{hc}\right)^4 \underbrace{\int_0^\infty \frac{z^3 dz}{e^z - 1}}_{\Gamma(3+1)\,\zeta(3+1) = \Gamma(4)\,\zeta(4) = \frac{\pi^4}{15}}$$

$$u = 8\pi \cdot \frac{\pi^4}{15} \frac{(k_B T)^4}{(hc)^3} = \frac{4\sigma}{c} T^4$$ (Stefan-Boltzman law)

$$\Rightarrow \sigma = \frac{2\pi^5}{15} \frac{k_B^4}{h^3 c^2}$$

From Wien's displacement law: $\frac{hc}{\lambda_m k_B T} = 5\left(1 - e^{-\frac{hc}{\lambda_m k_B T}}\right)$

$\Rightarrow$ solution $\frac{hc}{\lambda_m k_B T} = 4.965$.

But $\lambda_m T = 0.002899$ m·K $\Rightarrow$ we know $\frac{hc}{k_B}$!

$\Rightarrow$ Planck got $\frac{k_B^4}{h^3}$ from Stefan-Boltzmann law

and $\frac{h}{k_B}$ from Wien's displacement law

$\Rightarrow$ he got constant $h = 6.55 \cdot 10^{-27}$ erg · sec

erg $= 10^{-7}$ J

$$h = 6.55 \cdot 10^{-34} \text{ J} \cdot \text{s}$$

Modern value of Planck's constant:

$$\underline{h = 6.626 \cdot 10^{-34} \text{ J} \cdot \text{s}}$$

Planck's further insight:

spectral density $u(\lambda)d\lambda = \dfrac{8\pi hc \, d\lambda}{\lambda^5 (e^{hc/\lambda k_B T} - 1)}$

$$= \underbrace{u(\lambda, T)}_{\text{energy per mode}} \cdot \frac{D(\lambda)\, d\lambda}{\text{Volume}} = \frac{8\pi}{\lambda^4} d\lambda \cdot \frac{hc}{\lambda (e^{hc/\lambda k_B T} - 1)}$$

energy per mode ($= k_B T$ in Rayleigh-Jeans works)

$$\Rightarrow u(\lambda, T) = \frac{hc}{\lambda \left(e^{\frac{hc}{\lambda k_B T}} - 1\right)}$$

but $\frac{c}{\lambda} = \nu$ (frequency) $= \frac{\omega}{2\pi}$

$$\Rightarrow \frac{hc}{\lambda} = h\nu = \frac{h}{2\pi} \cdot \omega = \hbar\omega$$

$$u(\omega, T) = \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_B T}} - 1} \quad (**)$$

Planck observed that such a $U(\omega,T)$ obtains from the following entropy for 1 oscillator

$$\sigma_1 = \left(1+\frac{U}{\hbar\omega}\right)\ln\left(1+\frac{U}{\hbar\omega}\right) - \frac{U}{\hbar\omega}\ln\frac{U}{\hbar\omega}$$

$$\frac{1}{\tau} = \left(\frac{\partial\sigma}{\partial U}\right)_V = \left(\frac{\partial\sigma}{\partial x}\right)_V \frac{\partial x}{\partial U} = \left(\text{where } \frac{U}{\hbar\omega} = x \text{ for brevity}\right) =$$

$$= \frac{1}{\hbar\omega}\left(\frac{\partial\sigma}{\partial x}\right)_V$$

$$\frac{\hbar\omega}{\tau} = \frac{\partial}{\partial x}\left[(1+x)\ln(1+x) - x\ln x\right] = \frac{1+x}{1+x} + \ln(1+x) -$$

$$- \frac{x}{x} - \ln x = \ln\frac{1+x}{x}$$

$$\Rightarrow \quad \frac{1+x}{x} = e^{\hbar\omega/\tau} \quad \Rightarrow \quad 1+x = x e^{\hbar\omega/\tau}$$

$$x\left(e^{\frac{\hbar\omega}{\tau}}-1\right) = 1$$

$$x = \frac{1}{e^{\frac{\hbar\omega}{\tau}}-1}$$

$$\Rightarrow \quad U = \hbar\omega\cdot x = \frac{\hbar\omega}{e^{\frac{\hbar\omega}{\tau}}-1}$$

which is Planck's result (**)

But then $\sigma_1 = \left(1+\frac{U}{\hbar\omega}\right)\ln\left(1+\frac{U}{\hbar\omega}\right) - \frac{U}{\hbar\omega}\ln\frac{U}{\hbar\omega}$

is understood as coming from the entropy of N oscillators with total energy $U_N = M\cdot\hbar\omega = N\cdot U \Rightarrow \frac{M}{N} = \frac{U}{\hbar\omega}$

$$\Rightarrow \sigma_N = \ln g = \ln\frac{(N-1+M)!}{M!\,(N-1)!} = \cancel{(1+M)\ln(1+M)}$$

$$= (N+M)\ln(N+M) - M\ln M - N\ln N = (N+M)\ln\frac{N+M}{N} - M\ln M + M\ln N$$

$$= N\left\{\left(1+\frac{M}{N}\right)\ln\left(1+\frac{M}{N}\right) - \frac{M}{N}\ln\frac{M}{N}\right\} = N\left[\left(1+\frac{U}{\hbar\omega}\right)\ln\left(1+\frac{U}{\hbar\omega}\right) - \frac{U}{\hbar\omega}\ln\frac{U}{\hbar\omega}\right]$$

The meaning of this is: energy $U_N = N \cdot U$ of N oscillators (each of which has an average energy U) is represented by M quanta each of energy $\hbar\omega$.

Then $g = \dfrac{(N+M)!}{M!\,(N-M)!}$ is the number of ways to arrange M <u>indistinguishable</u> energy elements (quanta) among N oscillators.

The success of the calculation implies that energy of oscillator(s) is <u>quantized</u> in integer amounts of $\hbar\omega = h\nu$.

$$\boxed{\text{This is how Quantum Mechanics have entered physics in 1900.}}$$

Planck got quantization of energy from experimental data on $\dfrac{dQ}{dt}$ and their explanation.

[We consider] the distribution of the energy U among N oscillators of frequency ν. If U is viewed as divisible without limit, then an infinite number of distributions are possible. We consider however—and this is the essential point of the whole calculation—U as made up of an <u>entirely determined number of finite equal parts</u>, and we make use of the natural constant <u>$h = 6.55 \times 10^{-27}$ erg-sec.</u> This constant when multiplied by the common frequency of the oscillators gives the element of energy in ergs ...

Max Planck
1900

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modern value $h = 6.626 \cdot 10^{-27}$ erg·sec $=$
$= 6.626 \cdot 10^{-34}$ J·sec